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MgH_2 and LiH metal hydrides crystals as novel hydrogen storage material: Electronic structure and optical properties

A.H. Reshak [a,b,*]

[a] Institute of Complex Systems, FFPW, CENAKVA, University of South Bohemia in CB, Nove Hrady 37333, Czech Republic

[b] Center of Excellence Geopolymer and Green Technology, School of Material Engineering, University Malaysia Perlis, 01007 Kangar, Perlis, Malaysia

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ABSTRACT

We have performed a comprehensive theoretical investigation of the electronic band structure, density of states, electronic charge density and optical properties of the novel hydrogen storage material MgH_2 and LiH compounds. The all electron full potential linear augmented plane wave method was employed. The local density approximation (LDA), the generalized gradient approximation (GGA) and the Engle Vosko generalize gradient approximation (EVGGA) were used to treat the exchange-correlation potential. The calculations show that the MgH_2 compound is indirect gap semiconductor as the conduction band minimum (CBM) situated at R point of the Brillouin zone (BZ), while the valence band maximum (VBM) located between Λ and Γ points of the BZ, whereas LiH is a direct gap material as the CBM and the VBM located at X point of BZ. The values of the calculated energy band gap of MgH_2 (LiH) compounds are 3.372 (2.769), 3.735 (3.067) and 5.104 (4.488) eV for LDA, GGA, and EVGGA, respectively. From the partial density of states and the electronic charge density in (0 0 1) and (1 0 1) crystallographic planes we conclude that there exists strong ionic bonds. The bond lengths were calculated and compared with the available experimental and theoretical results, our results show better agreement with the experimental values than the other theoretical results. The frequency dependent dielectric function's dispersions were calculated and analyzed so as to obtain further insight into the electronic structure. The calculated dielectric function's dispersions confirm the semiconducting nature of MgH_2 and LiH compounds.

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1. Introduction

Hydrogen may be stored in molecular form as a liquid or compressed gas. Liquid hydrogen cannot be easy to maintain for long time without loss in the energy contents moreover, liquefaction process utilizes 30% of energy content of the stored hydrogen [1,2]. Compression also needs external energy, and acceptable storage densities require pressures in

* Institute of Complex Systems, FFPW, CENAKVA, University of South Bohemia in CB, Nove Hrady 37333, Czech Republic. Tel.: +420 777 729 583; fax: +420 386 361 219.
E-mail address: maalidph@yahoo.co.uk.

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excess of 350 bar, which pose engineering challenges. Molecular adsorption can partially alleviate the extreme pressures and temperatures associated with liquefied and compressed hydrogen [1,2]. However, because H_2 is nonpolar and contains only two strongly covalently bound electrons, electrostatic and dispersive bonds to adsorbents are weak. The best current materials, high surface area activated carbons, still require cryogenic temperature to achieve high (5.5 wt.%) storage densities [1,2]. Since hydrogen has high energy content 124 (MJ/kg), which is much higher than the other conventional sources of energy like petroleum (44 MJ/kg) [3–5]. This huge difference in energy content between both the fuels makes hydrogen energy to be able to use for automobiles. It is a challenging and important task to develop safe and efficient hydrogen storage systems in order to ensure hydrogen energy to be used practically. Many kinds of hydrogen storage materials have been explored, such as metal hydride systems [3–5], nanotubes [6–10], nanofibers [11–13], and metal-organic frameworks [14]. But so far metal hydrides are the leading hydrogen storage materials that are most widely studied and applied. However, the limited hydrogen contents of many metal hydrides (1.4 wt.% for $LaNi_5$, 1.8 wt.% for TiFe, and 3.6 wt.% for Mg_2Ni) have greatly affected the applications at commercial scale. The US Department of Energy (DOE) has set a hydrogen storage content goal of 6.5 wt.% for commercial applications [15]. Due to the great hydrogen capacities of LiH, MgH_2, and CaH_2 compounds, they are considered to be attractive metal hydrides and among them, MgH_2 (7.6 wt.% hydrogen content) have been extensively studied due to the weaker M–H bonding as compared to LiH and CaH_2 [16]. The consumption of hydrogen energy at large scale requires compact storage materials to be design with a high mass content of hydrogen [17]. One of the major disadvantages of metal hydrides for hydrogen storage is the low hydrogen weight capacity of these compounds which is due to the low atomic mass ratio of hydrogen to most of the metals and alloys. In addition hydrogen-storing material requires to have moderate stability (ability to release hydrogen in the ambient temperature range), fast kinetics of absorption, desorption and resistance towards hydrogenation poisons; these requirements preclude the use of light metals such as lithium or magnesium for hydrogen storage purposes [18]. Moreover, the formation energy of hydrogen storage materials has to be such that it is stable at room temperature, yet it has to decompose at low temperature to release its hydrogen [19]. Among the interesting hydrogen storage materials metallic hydrides have top priority due to high storage capacities at low pressure and also sustain volumetric densities as compared to liquid hydrogen [20]. Metallic hydrides have highest volumetric density of hydrogen [21]. Hydrogen storage in metal hydrides has been the focus of intensive research. Magnesium dihydride combines a high H_2 capacity with the benefit of the low cost of the abundantly available magnesium [22–24]. The main barriers for direct usage of pure MgH_2 are slow desorption kinetics, a high thermodynamic stability, and a high reactivity toward air and oxygen which it has in common with most other lightweight metal hydrides [25,26]. The high thermodynamic stability of MgH_2 results in a relatively high desorption enthalpy, which increases desorption temperature of 573 K at 1 bar H_2 [22,27,28]. Research efforts have been performed during the past two decades to modify the Mg-based system, aiming at increasing the absorption/desorption rates and lowering the desorption temperature. Different approaches were reported, mainly involving alloying Mg with other elements, high energy or reactive ball milling of Mg, or surface modification of Mg [29–36]. To make it possible for mobile applications in our daily life requires materials that have low decomposition temperature, high hydrogen densities and fast kinetics for absorption and desorption. Here we present first principles calculation of metal hydrides LiH and MgH_2 using the full potential linearized augmented plane waves (FP-LAPW) within three types of exchange-correlation potential.

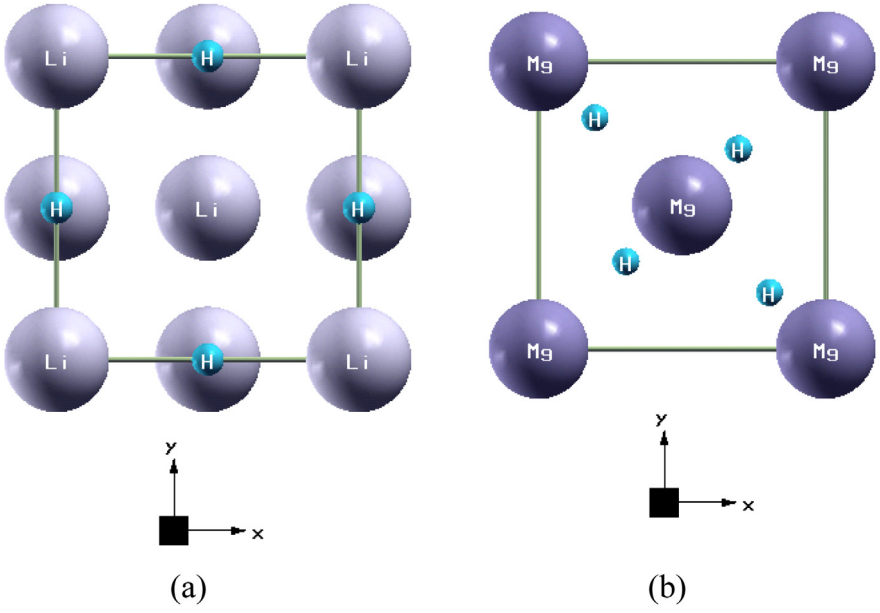


Fig. 1 – Unit cell structures for (a) LiH compound; (b) MgH_2 compound.

2. Computational method

We have carried out *ab-initio* calculations of the metal hydrides, MgH_2 and LiH as novel hydrogen storage material, using full potential linearized augmented plane wave (FP-LAPW) method within the frame work of density functional theory (DFT) as implemented in the WIEN2K code [37]. In this method the space is divided into an interstitial region and non-overlapping spheres centered at the atomic sites. In the present calculation $R_{MT}K_{max} = 7.0$ is used which determines matrix size (convergence), where K_{max} is the plane wave cut-off and R_{MT} is the smallest of all atomic radii. The self-consistent calculations are considered to be converged when the total energy of the system is stable within 10^{-4} Ry. The local density approximation (LDA) and the generalized gradient approximation (GGA) are used to treat the exchange-correlation potential [38]. In addition the Engle Vosko generalize gradient approximation (EVGGA) is used to avoid the underestimation of the energy band gap caused by LDA and GGA. Since EVGGA is producing better band splitting and more efficient in calculating the band structure and lead to enhance the band gap value [39–41]. Thus, we will show the results of EVGGA. The unit cell structures of LiH and MgH_2 are shown in Fig. 1a and b. LiH has face centered cubic (FCC) crystal structure, NaCl type-like structure. The atomic positions of Li and H atoms, are (0, 0, 0) and (0.5, 0.5, 0.5), respectively, and the

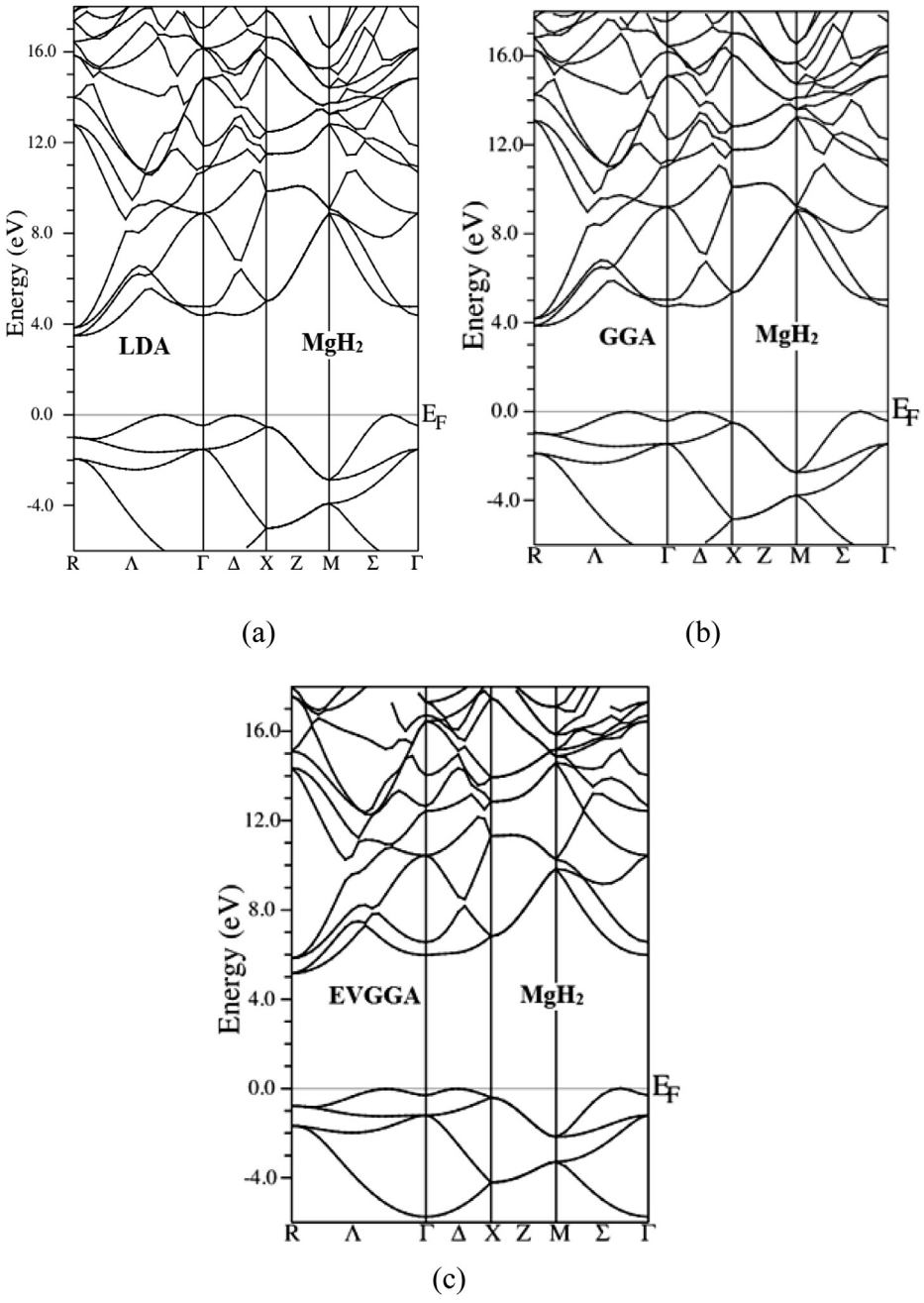


Fig. 2 – Calculated band structure of MgH_2 compound.

lattice parameter is 4.083 Å [42]. Metal hydride, MgH_2 crystallizes in the tetragonal structure with space group $P4_2/mnm$. The atomic positions of Mg and H atoms are (0, 0, 0) and (0.304, 0.304, 0), respectively. The experimental values of lattice constants, $a = 4.501$ Å and $c = 3.01$ Å are used for the calculation [43].

3. Results and discussion

3.1. *Band structure and density of states*

Fig. 2a–c shows the calculated band structure of MgH_2 using LDA, GGA and EVGGA respectively. While the band structure for LiH is shown in Fig. 3a–c. Following these figures one can see that for MgH_2 compound the conduction band minimum (CBM) situated at R point of the Brillouin zone (BZ), while the valence band maximum (VBM) is located between Λ and Γ points of the BZ resulting in indirect band gap. The value of the energy band gap of MgH_2 is increased from 3.372 eV for LDA to be 3.735 eV for GGA, while it increases dramatically with using EVGGA to be 5.104 eV, which is very close to the experimental value (5.16 eV) [48] and much better than the other theoretical findings which reported energy gap value around 3.60 eV [49], 3.4 eV [49,50] and 4.2 eV [51] for MgH_2. Whereas LiH compound exhibit direct band gap around X point of BZ. The value of the energy gap is about 2.769 eV for LDA, 3.067 eV for GGA, and 4.488 eV for EVGGA. The value of the energy gap obtained by

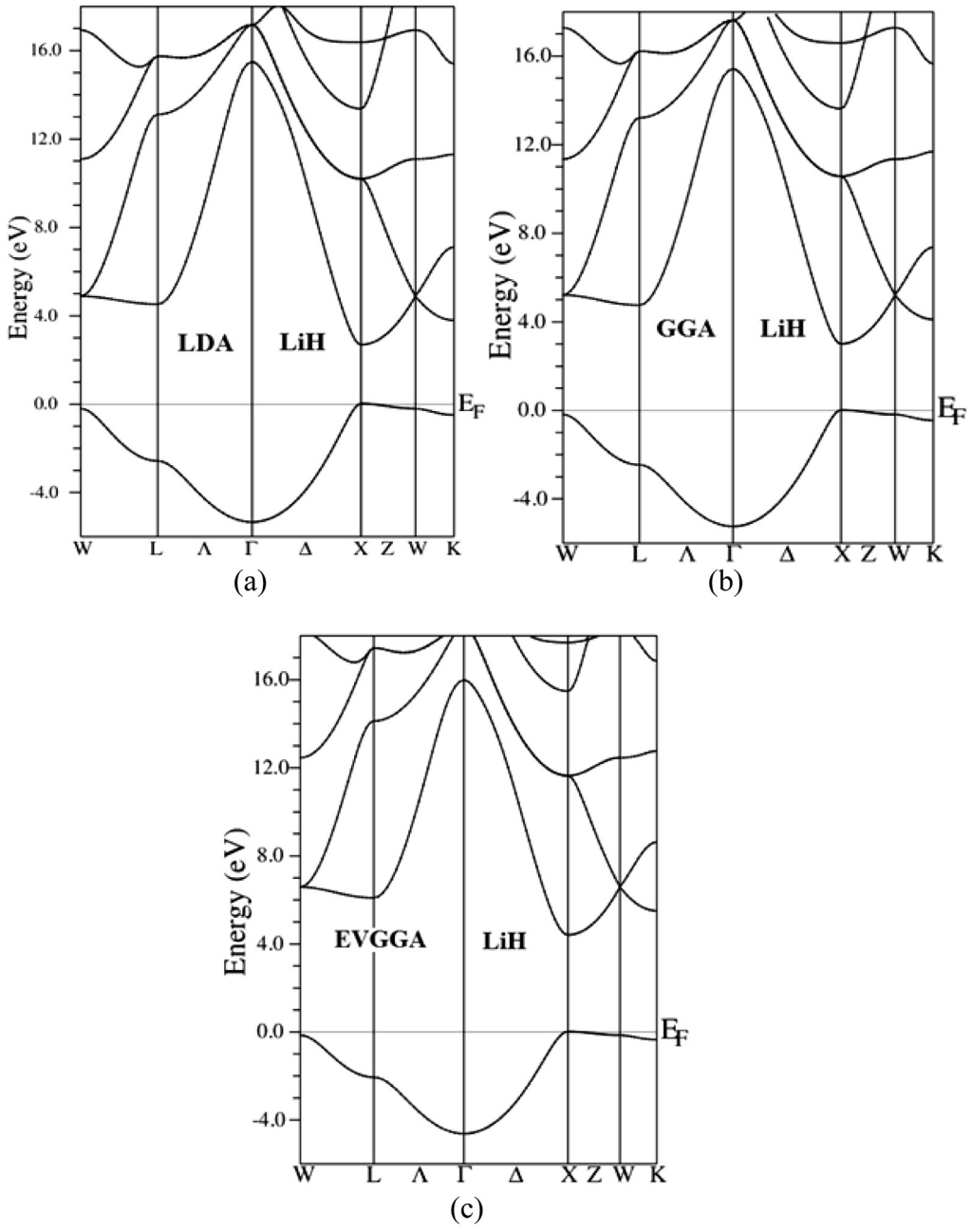


Fig. 3 – Calculated band structure of LiH compound.

EVGGA, shows very good agreement with the measured one 4.99 [44] and 5.0 eV [45–47] which is much better than the other theoretical value 3.2 eV [49]. The increasing in the energy band gap is attributed to the fact that with moving from LDA to GGA then EVGGA the CBM shift towards higher energies with respect to Fermi energy (E_F).

The density of electronic state enables us to analyze the relationship between electronic structure and bonding nature of the hydrogen storage materials. The total density of states (TDOS) and partial density of states (PDOS) for MgH_2 and LiH, metal hydrides, are shown in Figs. 4 and 5 respectively. The TDOS of MgH_2 is shown in Fig. 4a, in which the major bonding peak appears from −3.0 eV to E_F, and from the PDOS (Fig. 4b) it is revealed that this peak is dominated by the bonding of H-s, Mg-s, Mg-p electrons. The appearance of the wide energy gap confirm the non-metallic characteristic of MgH_2.

Fig. 5 shows the TDOS and PDOS of LiH. In the TDOS, the major bonding peak occurs between −2.0 eV and E_F, and from partials DOS (Fig. 5b) it can be seen that the bonding peak originate from the interaction of H-s and Li-s electrons. The existence of the wide energy gap confirm semiconducting nature of LiH. From the PDOS we can identify the various characters of the density of states; the structures at lowest energy are mainly due to H-s state with a small contribution from Mg-s and Mg-p states. The second structure/group between 5.1 eV and 6.0 eV consists of Mg-s, Mg-p and a small contribution from H-s state. While for LiH compound the structures at lowest energy is due to H-s state and small contribution from Li-s state. From E_F to 6.1 eV, Li-s state is dominant as compared to H-s state (see Fig. 5b).

From the calculated angular momentum decomposition of the atoms projected density of states one can elucidated the origin of the chemical bonding. From Figs. 4 and 5, it can be seen that the densities of states, extending from −5.0 eV up to E_F is mainly originated from H-s states (0.4 electrons/eV), Mg-p states (0.1 electrons/eV), and Mg-s states (0.09 electrons/eV) for MgH_2. While for LiH it is from H-s states (0.3 electrons/eV), and Li-s states (0.05 electrons/eV). This is obtained by taken very close looking at the angular momentum projected densities of states for both of MgH_2 and LiH compounds. Therefore these results show that there are some electrons transferred from H-s, Mg-s/p and Li-s states to valence bands and contribute in a strong ionic bonds. Also we can prove the ionic nature of the bonds in such a way, from PDOS it is clear that there is no hybridization between the states, except at very small energy regions, for example a deep look to the PDOS of MgH_2 compound one can notice that H-s states hybridize with Mg-s states between −5.0 and −4.0 eV, also Mg-p

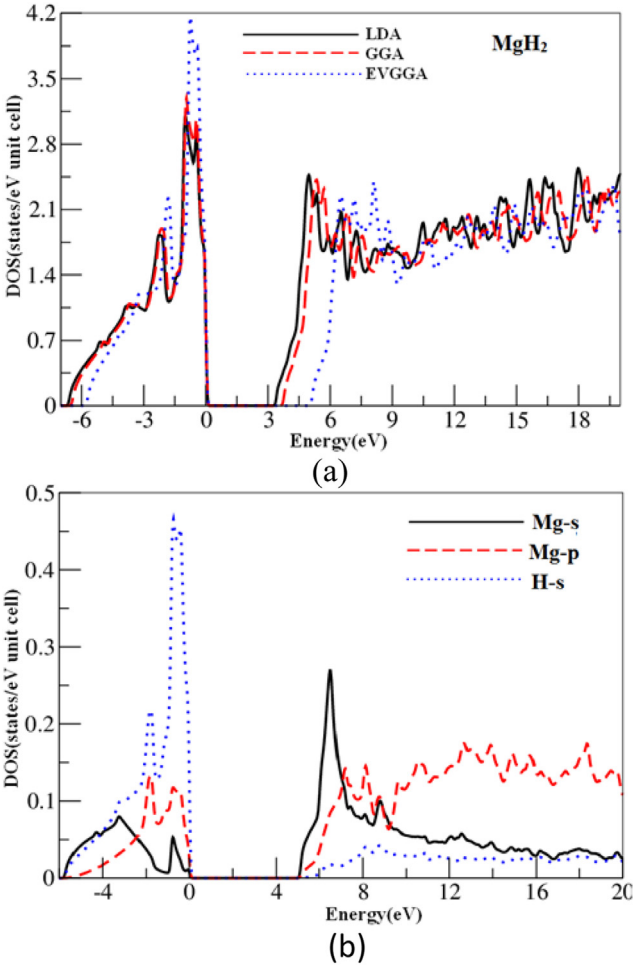


Fig. 4 – Calculated (a) Total density of states (States/eV unit cell) (b) PDOS for MgH_2.

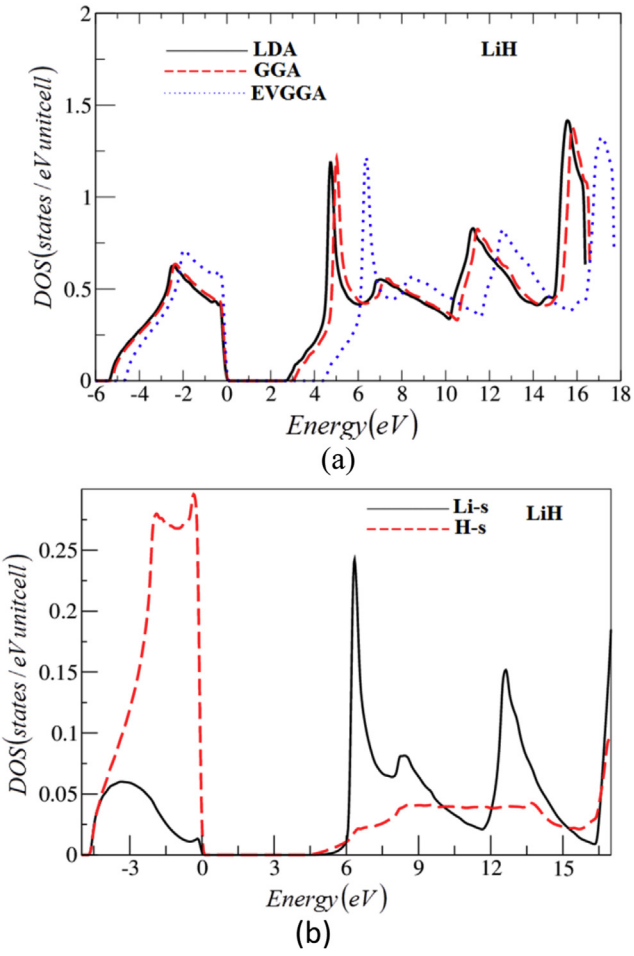


Fig. 5 – Calculated (a) Total density of states (States/eV unit cell) (b) PDOS for LiH.

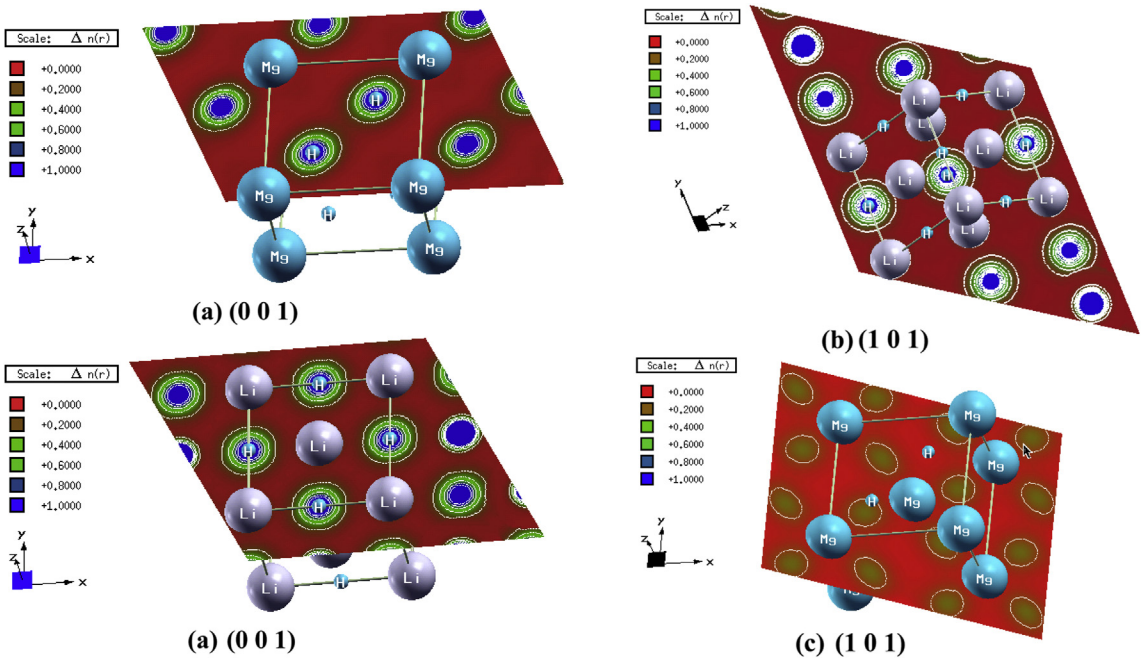


Fig. 6 – Electronic charge density (a) MgH_2 (b) LiH (c) LiH (d) MgH_2.

state hybridizes with Mg-s state at around 8.2 eV. Whereas for LiH compound Li-s state hybridizes with H-s state around −4.5 eV and 16.5 eV. In general the strong hybridization between the states exhibit strong covalent bonds. Hence, in the above cases there is very week hybridization between the states, thus there exists a strong ionic bonds.

3.2. *Electronic charge density*

Electronic charge density enables us to explore the nature of the chemical bonding between the atoms. Thus, we have calculated the electronic charge density to gain better insight of the chemical bonding and charge transfer of MgH_2 and LiH

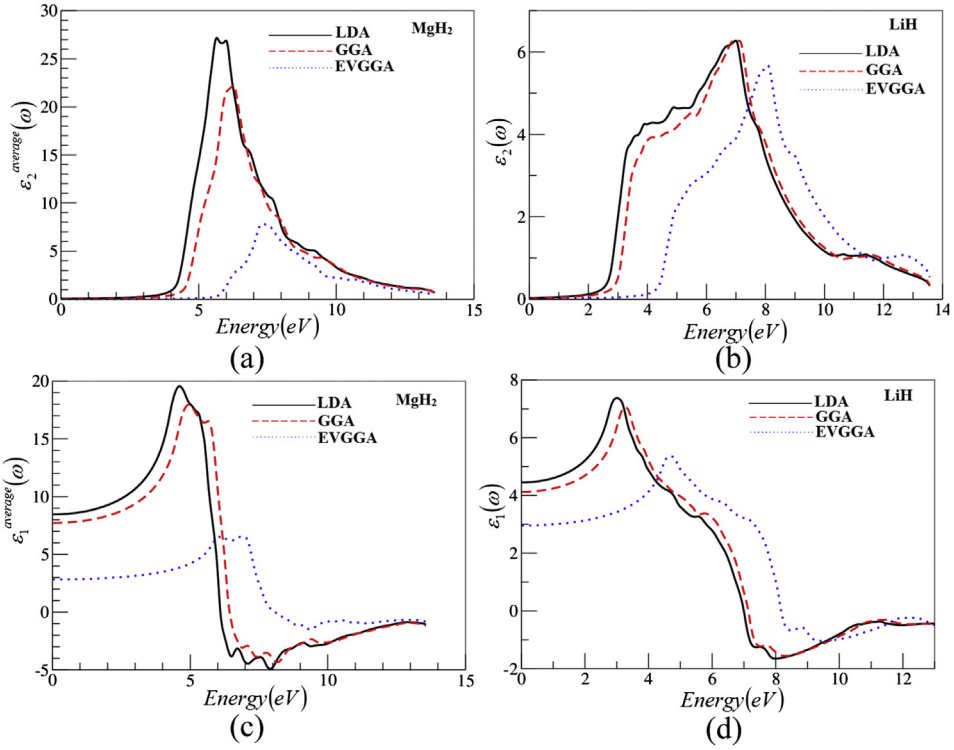


Fig. 7 – Calculated $\varepsilon_2^{average}(\omega)$, $\varepsilon_2(\omega)$, $\varepsilon_1^{average}(\omega)$ and $\varepsilon_1(\omega)$ for MgH_2 and LiH.

Table 1 – Calculated values of $\varepsilon_1^{average}(0)$, $\varepsilon_1^{xx}(0)$, $\varepsilon_1^{zz}(0)$ and $\varepsilon_1(0)$.

	LDA	GGA	EVGGA
MgH_2			
$\varepsilon_1^{average}(0)$	8.46	7.75	2.84
$\varepsilon_1^{xx}(0)$	4.24	3.86	2.84
$\varepsilon_1^{zz}(0)$	4.31	3.92	2.88
LiH			
$\varepsilon_1(0)$	4.45	4.12	2.96

compounds. The electro-negativity of H, Mg and Li is 2.2, 1.31 and 0.98, respectively. It is found that the majority of charges are accumulated on H site and the distribution of electronic charge is spherical which results in ionic bonding with Li and Mg (the Pauling electro-negativity between H and Mg is 0.89 and between H and Li is 1.22). H form polar ionic bonding with both of Li and Mg. Also according to the electro-negativity values of these atoms thus, the valence electrons from Li and Mg sites are transferred to H site. As the plot show that through the color charge density scale where blue color (+1.0000) corresponds to the maximum charge accumulation site. Fig. 6a and b, shows the electronic charge density plots in the (0 0 1) crystallographic plane for MgH_2 and LiH compounds, respectively. We have calculated bond lengths between Mg and H in MgH_2 and compare it with the available theoretical data [52,53]. Thus, our calculated bond lengths of Mg1–H1 is 1.935 Å, and Mg2–H1 is 1.955 Å, in comparison to the previous theoretical values (Mg–H1 = 1.919 Å, Mg–H2 = 1.919 Å) [52] and (Mg–H1 = 1.952 Å, Mg–H2 = 1.953 Å) [53]. Further insight we have calculated the electronic charge density in (1 0 1) crystallographic plane, as shown in Fig. 6c and d. This plane shows only the Mg atoms of MgH_2 compound in contrast to (0 0 1) crystallographic plane which shows the two atoms (Mg and H). While (1 0 1) crystallographic plane shows both atoms of LiH compounds similar to (0 0 1) crystallographic plane. These plots show there is a considerable anisotropy between the crystallographic planes (0 0 1) and (1 0 1) of MgH_2 compound. Whereas these plane show the isotropic nature of LiH compound.

3.3. *Optical properties*

The optical properties can provide detailed information about the electronic structure of the materials. The optical properties of solids are a major topic, both in basic research as well as for industrial applications. While for the former the origin and nature of different excitation processes are of fundamental interest, the latter can make use of them in many optoelectronic devices. We have calculated the frequency dependent dielectric function's dispersions of MgH_2 and LiH compounds. Since MgH_2 crystallizes in tetragonal symmetry, thus the symmetry allows only two nonzero tenser components. While only one nonzero tenser component allows for LiH as it crystallizes in a cubic symmetry. In order to seek the influence of using LDA, GGA and EVGGA one the optical properties thus the $\varepsilon_2^{average}(\omega)$ of MgH_2 is calculated and presented in Fig. 7a, while $\varepsilon_2(\omega)$ of LiH is calculated and presented in Fig. 7b. Following these figures, it is clear that all the structures of $\varepsilon_2^{average}(\omega)$ and $\varepsilon_2(\omega)$, is shifted towards higher energies with increasing the energy gap when moving from LDA to GGA to EVGGA. We should emphasize that LDA, GGA, and EVGGA

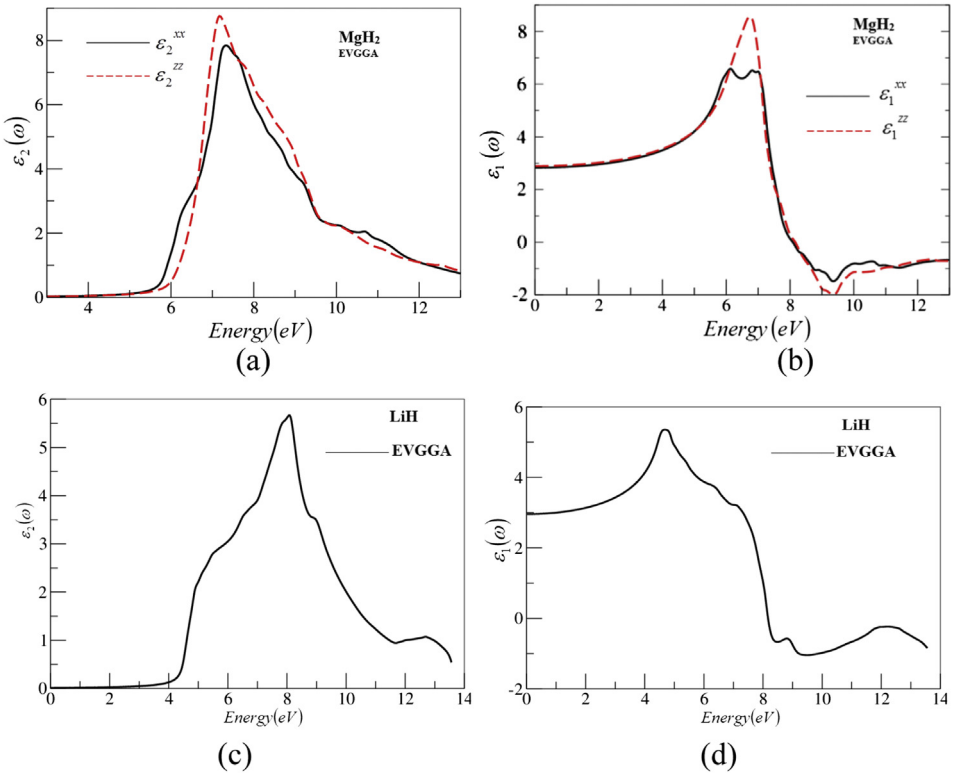


Fig. 8 – Calculated $\varepsilon_2^{xx}(\omega)$, $\varepsilon_2^{zz}(\omega)$ and $\varepsilon_2(\omega)$ and $\varepsilon_1^{xx}(\omega)$, $\varepsilon_1^{zz}(\omega)$ and $\varepsilon_1(\omega)$ for MgH_2 and LiH.

show the same general spectral features of $\varepsilon_2^{average}(\omega)$ and $\varepsilon_2(\omega)$ except the peak heights and positions, that is attributed to increase the value of the optical absorption edge (fundamental absorption edge) for, $\varepsilon_2^{average}(\omega)$ and $\varepsilon_2(\omega)$. These edges of optical absorption give the threshold for direct optical transitions between the top of the valence band and bottom of the conduction band. The values of the optical absorption edges of MgH_2 (LiH) compound are 3.372 (2.769), 3.735 (3.067) and 5.104 (4.488) eV for LDA, GGA, and EVGGA, respectively.

From the imaginary part of the dielectric function's dispersions the real part was calculated using Kramers–Kronig relations. The results of the calculated, $\varepsilon_1^{average}(\omega)$ and $\varepsilon_1(\omega)$ are shown in Fig. 7c and d. The calculated values of $\varepsilon_1^{average}(0)$ and $\varepsilon_1(0)$ are presented in Table 1.

Since EVGGA produce better band splitting and hence better energy gap's value, thus we will show more details of EVGGA results. Fig. 8a, illustrated the calculated $\varepsilon_2^{xx}(\omega)$ and $\varepsilon_2^{zz}(\omega)$ components of MgH_2 compound, there exists a considerable anisotropy between these components of the frequency dependent dielectric function. While $\varepsilon_2(\omega)$ of LiH is illustrated in Fig. 8c. The calculated $\varepsilon_1^{xx}(\omega)$, $\varepsilon_1^{zz}(\omega)$ and $\varepsilon_1(\omega)$ components of MgH_2 and LiH compounds are shown in Fig. 8b and d, and the corresponding values of $\varepsilon_1^{xx}(0)$, $\varepsilon_1^{zz}(0)$ and $\varepsilon_1(0)$ are listed in Table 1.

4. Conclusion

The all electron full potential linear augmented plane wave calculations are employed for calculating the electronic band structure, TDOS, PDOS, electronic charge density and the optical properties of the metal hydride, MgH_2 and LiH compounds. Calculations were performed using three types of exchange-correlation potential, namely LDA, GGA, and EVGGA. The electronic band structure of MgH_2 compound shows the CBM is situated at R point of BZ, while the VBM located between Λ and at Γ points of the BZ resulting in indirect band gap. The value of the energy band gap of MgH_2 is increased from 3.372 eV for LDA to be 3.735 eV for GGA, while it increases dramatically with using EVGGA to be 5.104 eV. Whereas LiH compound exhibits direct band gap around X point of BZ. The value of the energy gap is about 2.769 eV for LDA, 3.067 eV for GGA, and 4.488 eV for EVGGA. The increasing in the energy band gap is attributed to the fact that with moving from LDA to GGA then EVGGA the CBM shift towards higher energies with respect to E_F. Our calculated band gap value (5.104 eV) for MgH_2 using EVGGA is very close to the experimental value (5.16 eV) [48] and much better than the other theoretical findings which reported energy gap value around 3.60 eV [46], 3.4 eV [46,47] and 4.2 eV [48]. Whereas for LiH our calculated band gap (4.488 eV) shows very good agreement with the experimental value (4.99 eV) [41] and 5.0 eV [42–44] which is much better than the other theoretical value 3.2 eV [46]. From the PDOS and the electronic charge density in (0 0 1) and (1 0 1) crystallographic planes we conclude that there exists strong ionic bonds. The frequency dependent dielectric function's dispersions were calculated and analyzed. The optical properties confirm the semiconducting nature of MgH_2 and LiH compounds.

Acknowledgments

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